A Skew-Tensor Obstruction to Removing Symmetry in a Bourgain–Brezis Estimate

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Source Problem

The source is Jean Van Schaftingen, *Limiting Bourgain-Brezis estimates for systems: theme and variations*, arXiv:1311.6624 [1]. In Section 3.5, after stating the higher-order endpoint estimate on nilpotent homogeneous groups for symmetric horizontal k-tensors, the paper says:

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and if $\varphi \in C_c^\infty(G; \mathrm{Sym}^k(T_b G))$ *is a section, then*

$$\left| \int_G \langle \varphi, f \rangle \right| \leq C \|f\|_{\mathrm{L}^1(G)} \|\nabla_b \varphi\|_{\mathrm{L}^Q(G)} .$$

Here $\mathrm{Sym}^k(T_b G)$ denotes the bundle of symmetric k-linear maps on the horizontal bundle. It is not known whether the symmetry assumption is necessary for this inequality to hold [34, open problem 1]. This result was used to obtain the Gagliardo–Nirenberg inequality for forms in the Rumin complex [78–81] for forms on the Heisenberg groups $\mathbb{H}^1$ and $\mathbb{H}^2$ [10]. Since the Rumin complex contains higher-order differential operators, the higher-order estimates play a crucial role in the proof. The method would require heavy explicit computations of complexes to be generalized to higher-order Heisenberg groups.

The counterparts of the stronger theorems 2 and 4 have been obtained by Yi Wang and Po-Lam Yung [110] following the strategy of Bourgain and Brezis [16].

This packet answers the literal symmetry-removal question negatively: the displayed estimate cannot hold if φ is allowed to be an arbitrary horizontal k-tensor. The obstruction already occurs for $G = \mathbb{R}^2$, viewed as a step-one stratified group, and $k = 2$.

Result

Theorem 1. *The higher-order estimate from Section 3.5 of the source paper becomes false if the test tensor* $\varphi \in C_c^\infty(G; \mathrm{Sym}^k(T_b G))$ *is replaced by an arbitrary tensor* $\varphi \in C_c^\infty(G; \otimes^k T_b G)$*. In fact, it is false already for the Euclidean group* $G = \mathbb{R}^2$*, homogeneous dimension* $Q = 2$*, and* $k = 2$*.*

Proof Intuition

The source constraint pairs f against second horizontal derivatives of scalar functions. In Euclidean space these second derivatives are Hessians, hence symmetric bilinear forms. Therefore a purely antisymmetric tensor field automatically satisfies the constraint. If the test tensor were also allowed to be antisymmetric, the estimate would have to control the value of a critical Sobolev function at a point by its L^2 gradient norm. Logarithmic cutoffs show that such point control is impossible.

Proof

Proof. Work on $G = \mathbb{R}^2$, whose horizontal bundle is the whole tangent bundle and whose homogeneous dimension is $Q = 2$. Let

$$A = e_1 \otimes e_2 - e_2 \otimes e_1$$

be the standard nonzero antisymmetric 2-tensor.

Fix a nonnegative $\rho \in C_c^\infty(B(0,1))$ with $\int_{\mathbb{R}^2} \rho = 1$. For $m \geq 1$, set $r_m = e^{-2m}$ and

$$\rho_m(x) = r_m^{-2}\rho(x/r_m), \qquad f_m(x) = \rho_m(x)A.$$

Then $f_m \in C_c^\infty(\mathbb{R}^2; \otimes^2\mathbb{R}^2)$ and

$$\|f_m\|_{L^1} = |A| \int_{\mathbb{R}^2} \rho_m = |A|.$$

Moreover f_m satisfies the annihilation hypothesis from the higher-order estimate. Indeed, for every $\psi \in C_c^\infty(\mathbb{R}^2)$, the Hessian $D^2\psi(x)$ is symmetric, and hence

$$\langle D^2\psi(x), A\rangle = 0 \qquad \text{for every } x \in \mathbb{R}^2.$$

Therefore

$$\int_{\mathbb{R}^2} \langle D^2\psi, f_m\rangle = 0.$$

Assume, toward a contradiction, that the source estimate remained true with arbitrary 2-tensor test fields. Then there would be a constant C such that, for all such f satisfying the above annihilation condition and all $\Phi \in C_c^\infty(\mathbb{R}^2; \otimes^2\mathbb{R}^2)$,

$$\left| \int_{\mathbb{R}^2} \langle \Phi, f\rangle \right| \leq C\|f\|_{L^1(\mathbb{R}^2)} \|\nabla\Phi\|_{L^2(\mathbb{R}^2)}. \tag{1}$$

Choose smooth radial cutoffs $h_m \in C_c^\infty(B(0,2))$ with

$$h_m = 1 \quad \text{on } B(0, e^{-m})$$

and

$$\|\nabla h_m\|_{L^2(\mathbb{R}^2)} \leq C_0 m^{-1/2}.$$

For example, smooth a standard logarithmic cutoff that equals 1 for $r \leq e^{-m}$, equals $\log(1/r)/m$ for $e^{-m} < r < 1$, and equals 0 for $r \geq 1$. The estimate follows from

$$\int_{e^{-m}<|x|<1} \frac{1}{m^2|x|^2}\, dx = \frac{2\pi}{m}.$$

Now set $\Phi_m = h_m A$. Since $r_m = e^{-2m} < e^{-m}$, we have $h_m = 1$ on $\operatorname{supp} \rho_m$. Hence

$$\left| \int_{\mathbb{R}^2} \langle \Phi_m, f_m \rangle \right| = \left| \int_{\mathbb{R}^2} h_m(x) \rho_m(x) \langle A, A \rangle \, dx \right| = |A|^2.$$

On the other hand,

$$\|\nabla \Phi_m\|_{L^2} = |A| \|\nabla h_m\|_{L^2} \le |A| C_0 m^{-1/2}.$$

Putting $f = f_m$ and $\Phi = \Phi_m$ in (1) gives

$$|A|^2 \le C |A|^2 C_0 m^{-1/2}$$

for every m, which is impossible as $m \to \infty$. Thus the estimate cannot hold without the symmetry restriction on Φ. □

Limitations

This is a counterexample to the naive or literal removal of the symmetry restriction on the test tensor. It does not rule out repaired variants that add a separate condition annihilating or controlling antisymmetric components. Nor does it settle the Besov and Sobolev–Lorentz open problems stated earlier in the same source paper.

Novelty and Verification Notes

The local registry, solution index, attempt index, and proof-gap index were searched for arXiv:1311.6624 and for the phrases "symmetry assumption", "symmetric tensor", "higher-order group estimate", "Chanillo Van Schaftingen", and "antisymmetric". No local duplicate was found; the only "antisymmetric" hits were unrelated martingale and operator-space packets. Web searches on 2026-07-18 for the exact source sentence and close variants found the source paper or bibliographic mirrors, but no explicit counterexample to the literal symmetry-removal extension.

For human review, the main interpretation issue is whether the source's question means precisely replacing $\mathrm{Sym}^k(T_b G)$ by the full tensor bundle in the displayed estimate. Under that reading, the proof above is a direct counterexample.

References

[1] Jean Van Schaftingen, *Limiting Bourgain-Brezis estimates for systems: theme and variations*, arXiv:1311.6624, 2013; J. Fixed Point Theory Appl. 15 (2014), no. 2, 273–297.